

4 OPERATION PROCEDURES

4.1 Summary

This chapter describes emergency stop procedures, production outline, production procedures, and each operation during production.

**WARNING**

- **Thoroughly understand the contents of "3 OPERATING PANELS" before operating the device.**
-

<Contents>

- Outline of the production, preparation work, and actual procedures
- Basic adjustment procedures during production

<Purpose>

- To understand and master the basic procedures for daily production and the basic adjustment procedures during production.

<Intended reader>

- Operators
- System administrators

4.2 Emergency Stop and Restart

Perform the following procedures when stopping the device in the event of an emergency and then restarting it.

Emergency stop procedures

1. Turn OFF the main power switch on the side of the remote control unit.
 - ▶ Production is terminated.
 - ▶ The display of the control panel disappears.

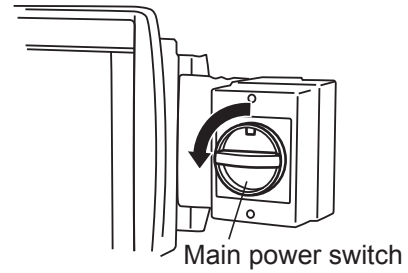


Fig. 4-1

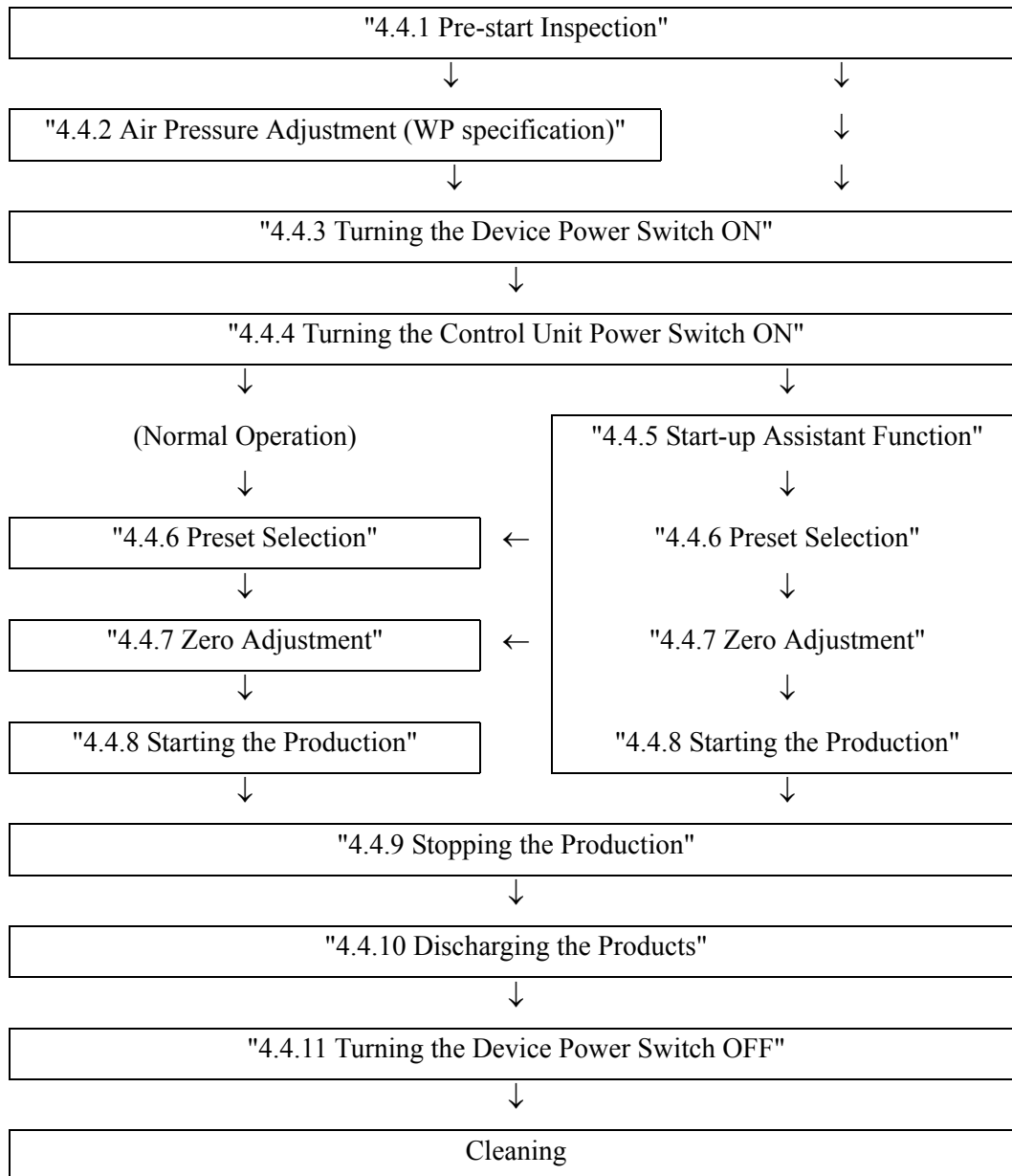
Restart procedures

1. Eliminate the cause of the emergency stop.
2. Restart the production using normal procedures.

4.3 Outline of the Production

This section describes the outline of the production.

Refer to "4.4 Production Procedures" and the subsequent sections for details regarding production procedures.



4.4 Production Procedures

This section describes the production procedures.

4.4.1 Pre-start Inspection

CAUTION

- For maintenance and inspections, unless instructed, the operator must turn OFF and lock the main power switch, and keep the key in his possession during the work (Refer to "1.6 Power Supply Shut Off and Indication").
- If any part of the device is not securely installed, it may fall due to the vibration during the production and may damage the device or injure personnel.
- If any problem is found during the pre-start inspection, contact the maintenance engineer and do not start the operation until the problem is solved.

Perform the following pre-start inspection before starting the production to eliminate any cause of the trouble.

1. Make sure that tools and any other irrelevant objects are not placed on top or in the vicinity of the device.
2. Make sure that the dispersion table is securely installed with no play.
3. Make sure that the dispersion table does not make contact with the radial troughs.
4. Make sure that the radial troughs do not make contact with each other.
5. Make sure that the pool hopper, weigh hopper and timing hopper (optional) are securely installed with no play.
6. Make sure that the collection chute is securely installed.
7. Make sure that the timing hopper and the drive unit are securely connected with the drive shaft (Optional).

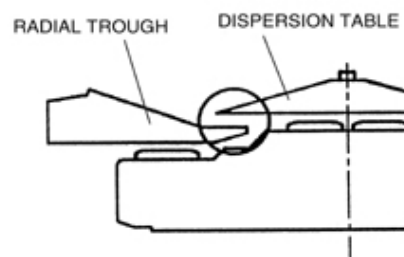


Fig. 4-2

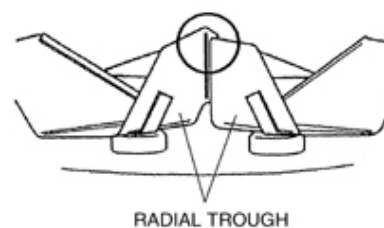


Fig. 4-3

4.4.2 Air Pressure Adjustment (WP specification)

1. Shut the air supply cockstop to stop the compressed air supply.
2. Turn the drain to let the remaining air out of the air dryer.

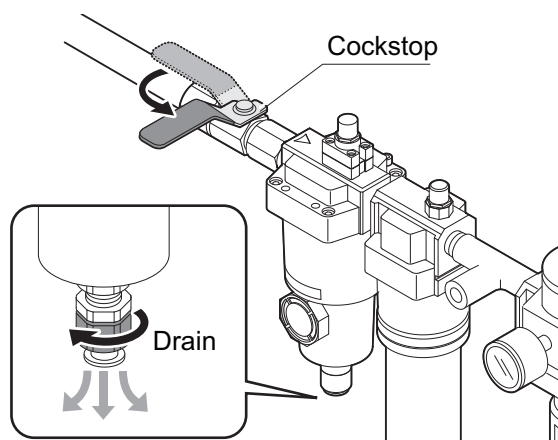


Fig. 4-4

3. Check that the pressure gauge reads 0Mpa.
4. Re-tighten the drain and open the air supply cockstop.
5. Check that the pressure gauge reads between 0.01 Mpa and 0.02 Mpa.

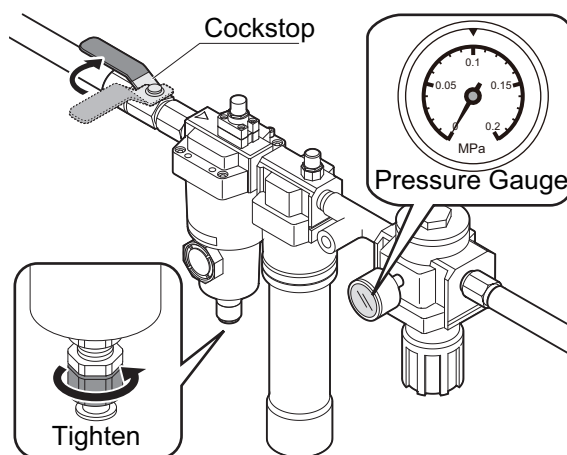


Fig. 4-5

- If the gauge does not read between 0.01 Mpa and 0.02 Mpa, adjust the gauge by turning the pressure adjustment knob to the right or left.

CAUTION

- When the pressure gauge does not read 0.02 Mpa or under, even after manually adjusting, check the pressure supply.
- Do not set the air pressure is too high as this may prevent accurate weighing.

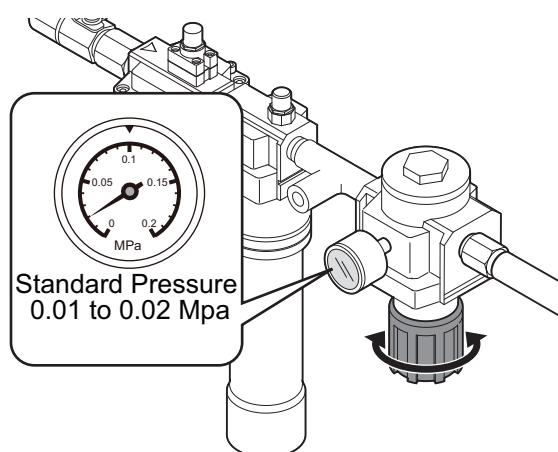


Fig. 4-6

4.4.3 Turning the Device Power Switch ON

CAUTION

- Turn the main power switch ON at least 30 minutes before starting the production in order to get stable weighing results.

This section describes how to supply power to the device.

1. Turn the main power switch ON.
 - ▶ The device is powered on.
2. Wait at least 30 minutes.

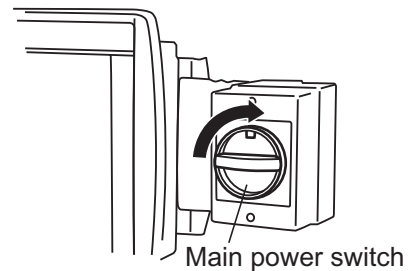


Fig. 4-7

4.4.4 Turning the Control Unit Power Switch ON

This section describes how to supply power to the device control unit.

1. Press the [Power] key.
 - ▶ The device control unit is powered on.



Fig. 4-8

4.4.5 Start-up Assistant Function

The start-up assistant function guides you through the steps up to the production and simplifies them by following the instructions on the screen.

When not using the start-up assistant function, proceed to "4.4.6 Preset Selection".

1. Press the [Start-up Assistant] key.

► The Start-Up Assistant screen appears.



Fig. 4-9

2. To cancel the start-up assistant function and return to the Main Menu screen, press the [Cancel] key.

Press the [Next] key to continue.

► The Select Preset screen of the start-up assistant menu appears.

The key for the next operation is blinking.

The outline of the operation performed via the start-up assistant function is displayed on the left side of the screen.

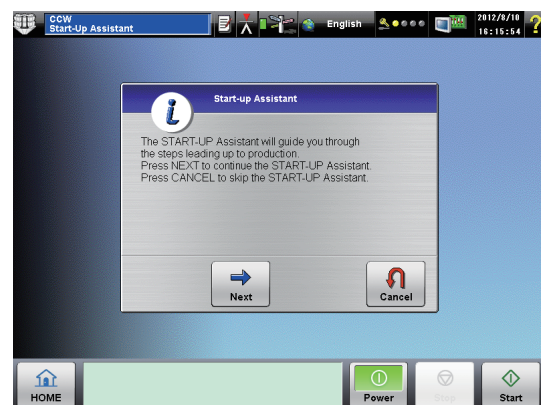


Fig. 4-10

3. Scroll the screen by using the scroll key or the scroll bar until the desired preset item is displayed.

4. Press the [Select Preset] key to select.

5. Press the [OK] key.

► The WH Zero Adjustment Start screen of the start-up assistant menu appears.



Fig. 4-11

NOTE

- On the WH Zero Adjustment screen, the weigh hoppers are displayed in blue. However, they are displayed in red right after power-on or when the zero adjustment is not performed for 30 minutes or more. When the weigh hoppers are displayed in red, it is not possible to proceed to the next step until the zero adjustment is performed.

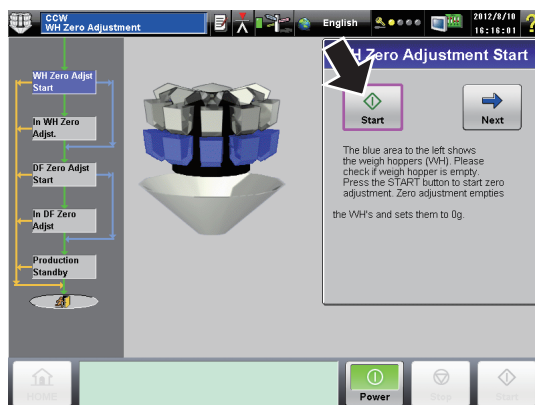


Fig. 4-12

6. Press the [Start] key.

- Zero adjustment operation for the weigh hopper starts. The progress bar is displayed during zero adjustment.

NOTE

- When an error occurs during zero adjustment, the weigh hoppers are displayed in red. In this case, press the [Start] key again and perform zero adjustment.

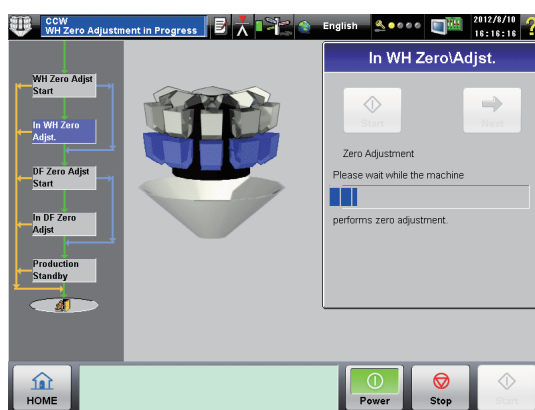


Fig. 4-13

- The DF Zero Adjustment Start screen of the start-up assistant menu appears.

NOTE

- On the DF Zero Adjustment screen, the dispersion table is displayed in blue. When the dispersion table is displayed in red immediately after power-on, it is not possible to proceed to the next step until the zero adjustment is performed.

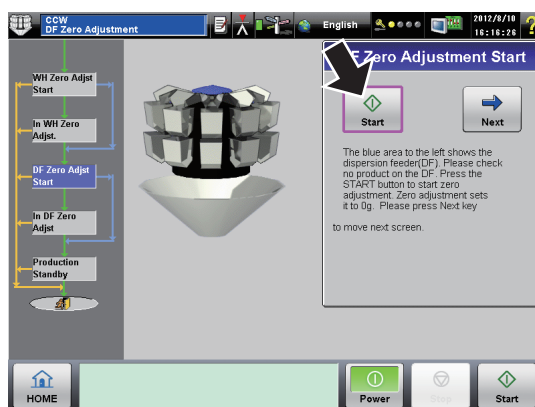


Fig. 4-14

7. Press the [Start] key.

- ▶ Zero adjustment operation for the dispersion table starts.
- ▶ When the zero adjustment operation for the dispersion table finishes without error, the Production Standby screen appears. The progress bar is displayed during zero adjustment.

NOTE

- When an error occurs during zero adjustment, the dispersion table is displayed in red. In this case, press the [Start] key again and perform zero adjustment.

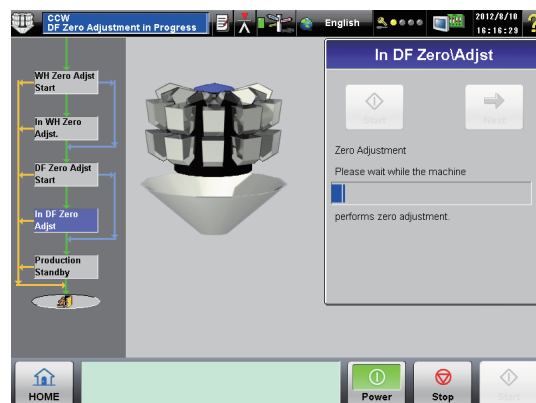


Fig. 4-15

8. Press the [Start] key.

- ▶ The production starts.

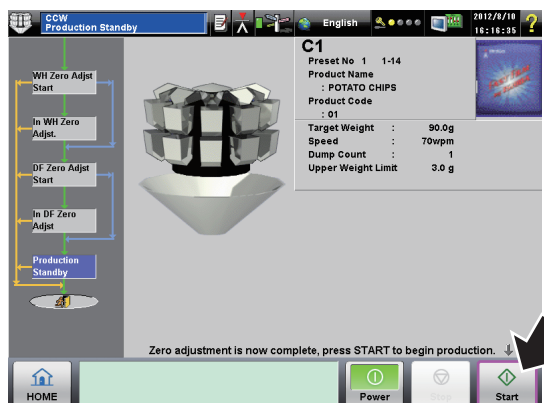


Fig. 4-16

TIP

- The start-up assistant function ends when the production starts and the normal operation menu appears.
- The start-up assistant function covers the steps up to "4.4.8 Starting the Production".

NOTE

- The start-up assistant function is available at any time, but the appearance of the screen differs depending on the machine condition just before activating the function.

4.4.6 Preset Selection

Select and use the preset item to which setting information of the product is registered in advance. This function makes it easy to set the product information.

1. On the Main Menu screen, press the photo displayed in the preset detail information area.

► The Select Preset screen appears.



Fig. 4-17

2. Use the scroll key or the scroll bar and press the desired [Preset Item] key.

3. Press the [Exit] key.

► The display returns to the Main Menu screen with the selected preset information set.



Fig. 4-18

<Switching the display for the Select Preset screen>

There are two display modes in the Select Preset screen: the photo mode and the list mode. In the photo mode, you can select the product visually; in the list mode, you can sort the items by product category, weight, etc.

1. Press the [Slct Dsply] key on the Select Preset screen.

► The display switches to the list mode.



Fig. 4-19

NOTE

- Pressing the title of each column ("PresetNo", "PdctClas", "Product Name", "Target Wt", "Speed", "Dump Cnt", and "Record Time") will sort the preset data by the selected column. For example, when the [Target Wt] key is pressed, the preset data is displayed in ascending order by target weight.

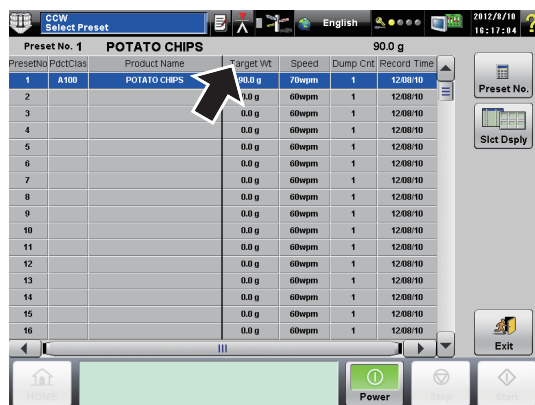


Fig. 4-20

2. Press the [Slct Dsply] key.

► The display returns to the photo mode.



Fig. 4-21

<Selecting the preset number directly>

The preset number can be directly selected using the Numeric Keypad screen. This function is useful when the preset number is already identified and the number is too large to find by scrolling through the screen.

1. Press the [Preset No.] key on the Select Preset screen.
 - ▶ The Numeric Keypad screen appears.
2. Input the desired preset number.
 - ▶ The Numeric Keypad screen disappears, and the entered preset number is selected.



Fig. 4-22

4.4.7 Zero Adjustment

Perform zero adjustment for the weigh hopper and dispersion table in order to weigh the product precisely.

<Zero adjustment for the weigh hopper>

1. Press the [Zero Adjst] key.
 - ▶ The Zero Adjustment screen appears.
 - ▶ All hoppers are selected and displayed in blue.



Fig. 4-23

2. Press the [Start] key.

NOTE

- The [Start] key cannot be pressed when the control unit power switch is OFF.



Fig. 4-24

- The message "Please wait a moment." appears on the screen, and zero adjustment starts.

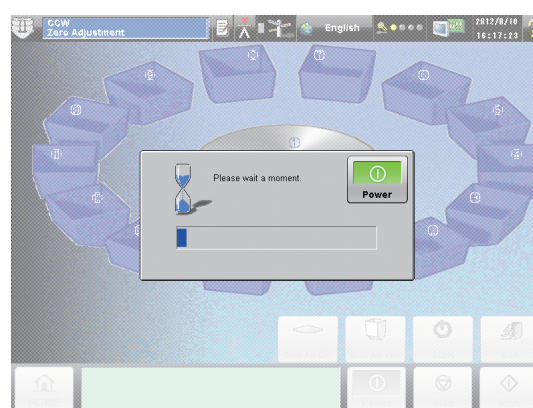


Fig. 4-25

- When zero adjustment operation finishes, each [WH] key displays the respective weight.
3. Make sure that each weight display of the [WH] key is within 0.0 ± 0.1 g.

NOTE

- If the weight display reading exceeds 0.1 g or falls below -0.1 g, perform the zero adjustment again.
- Zero adjustment may not be performed accurately within 30 minutes after turning the power on or when there is air movement around the machine.
- Pressing each [Weigh Hopper] key on the screen can switch the hopper status to selected or deselected.



Fig. 4-26

Next, perform the zero adjustment operation for the dispersion table.

<Zero adjustment for the dispersion table>

1. In the Zero Adjustment screen, press the [Slct All DF] key.

► All dispersion tables are displayed in blue.

2. Make sure that there is no product on the dispersion table.

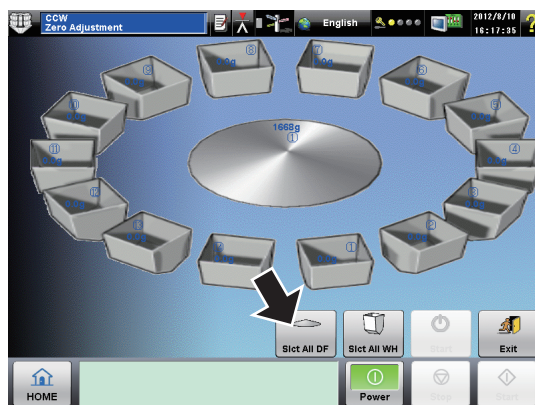


Fig. 4-27

3. Press the [Start] key.

NOTE

- The [Start] key cannot be pressed when the control unit power switch is OFF.



Fig. 4-28

- The message "Please wait a moment." appears on the screen, and zero adjustment starts.

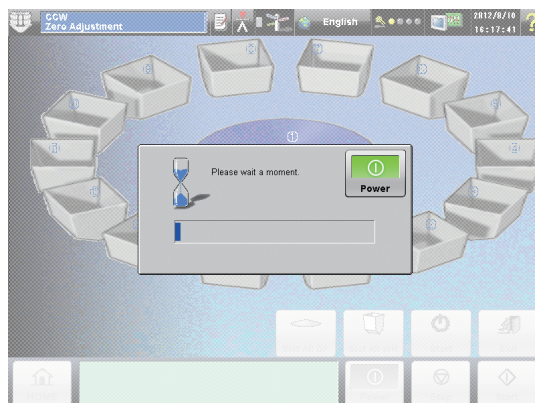


Fig. 4-29

► When zero adjustment operation finishes, each [DF] key displays the respective weight.

4. Make sure that each weight display of the [DF] key is 0 g.

NOTE

- When the weight display reading exceeds 0 g or falls below 0 g, perform the zero adjustment again.

5. Press the [Exit] key.

► The Main Menu appears.



Fig. 4-30

4.4.8 Starting the Production

Start the production.

1. Press the [Start] key.

► The production starts and the Production screen appears.

TIP

- When the feeder is interlocked with this device to automatically supply the product, pressing the [Infeed Control] key to ON will start the product supply.
- Refer to "6 FUNCTIONS OF THE OPERATION SCREENS" for each symbol displayed on the Production screen.
- Refer to "4.6 Temporary Stop and Restart" for temporarily stop of the production.



Fig. 4-31

CAUTION

- **DO NOT, UNDER ANY CIRCUMSTANCES, CLOSE THE HOPPERS WITH HANDS OR TOOLS.** Doing so may damage the drive unit.
- **When the main power switch is turned OFF while the hoppers are open, or when the hoppers remain open due to an electrical power failure, follow the procedures described in "4.9 Handling Drive Unit" for the corrective action.**

4.4.9 Stopping the Production

Stop the production.

1. When the infeed control is on, press the [Infeed Control] key.

► The lamp of the [Infeed Control] key turns off and the product supply to the device is terminated.



Fig. 4-32

2. Press the [Stop] key.

► The production is terminated.



Fig. 4-33

3. Press the [Exit] key.

► The display returns to the Main Menu.

NOTE

- To discharge the product or change the preset item after stopping the production, press the [Exit] key to return to the Main Menu before performing any operation.



Fig. 4-34

4.4.10 Discharging the Products

Discharge the remaining products from the device.

<Discharging the products>

1. Press the [Drain] key.
 - ▶ The Drain screen appears.
 - ▶ Discharging of the product starts.



Fig. 4-35

2. After discharging all products, press the [Drain STOP] key.
 - ▶ Discharging is terminated.

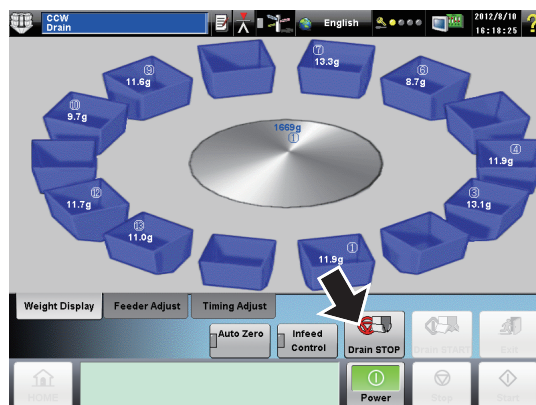


Fig. 4-36

3. Press the [Exit] key.
 - ▶ The display returns to the Main Menu.



Fig. 4-37

4.4.11 Turning the Device Power Switch OFF

Shut off the power supplied to the device.

1. Press the [Power] key.

- ▶ The control system power supply is turned OFF.



Fig. 4-38

2. Turn the main power switch OFF.

- ▶ The display of the control panel disappears.
- ▶ The power supplied to the device is shut off.

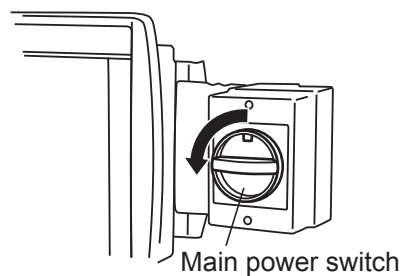


Fig. 4-39

<Cleaning>

Clean the device after shutdown.

4.5 Display During the Operation

The screen display during the operation can be switched by selecting a tab from the following: Combination, Feeder Adjust, Timing Adjust, Total Data, and Weight Display (Refer to "6.6 Production screen").

In the Combination screen, the display can be switched between the combination display and the expansion display, by using the [Select Display] pop-up key (Refer to "4.5.1.1 Select Display Pop-up Menu in Combination Screen").

In the combination display, each head weight, head status and preset information in addition to the combination weight value are displayed. In the expansion display, only the combination weight value is displayed.

- Combination weighing (Refer to "4.5.1 Combination Screen")
For a detailed description of the following functions, refer to the respective section.
- Feeder adjustment (Refer to "6.12 Feeder Adjustment Screen")
- Timing adjustment (Refer to "6.13 Timing Adjustment Screen")
- Total data (Refer to "6.11 Select Total Pop-up Menu")
- Weight display (Refer to "6.6.5 Weight Display Tab Screen")

4.5.1 Combination Screen

To display the Combination screen, press the [Combination] tab on the Production screen.



Fig. 4-40

Table 4-1

No.	Name	Function
1	Combination weight display	The combined weight value is displayed on each channel. The lamp indication varies depending on the results. Green lamp: Proper weight Yellow lamp: Over weight Red lamp: Under weight
2	Head condition display	Displays the combination and weighing status. ①: Head which was weighed and is stable. ①: Head which was weighed, was selected for combination and discharged products. The color of the circle indicates the head stability, and there are three levels: green (most stable), white (more stable) and yellow (stable). Empty head. Blank: Unstable head. Head under the auto zero adjustment. Head with error of overload or jam. Head that is set to be deactivated.
3	Head display guide	Displays the scrolling messages for the descriptions of head symbols.
4	Preset selection display	Displays the selected preset.
5	[Select Display] pop-up key	Selects the display method from either the combination or expansion display. Combination and expansion display icons are displayed by pressing this key. Pressing either icon will switch the display to the selected method.

4.5.1.1 Select Display Pop-up Menu in Combination Screen

On the Combination screen, either the "Combination" or "Expansion Display" mode can be selected. Press the [Select Display] pop-up key on the Combination screen, and select the desired display mode.

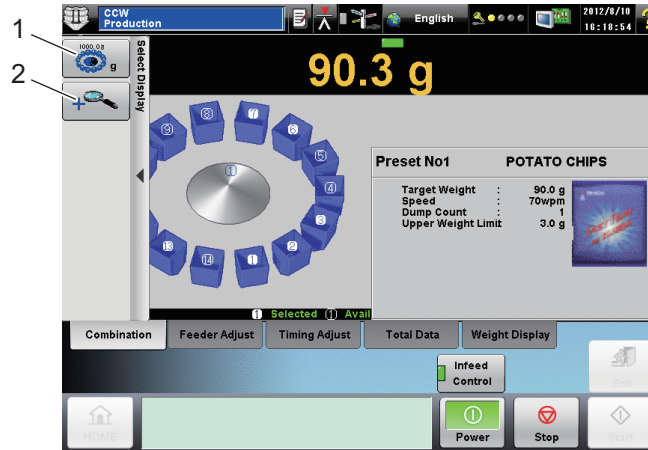
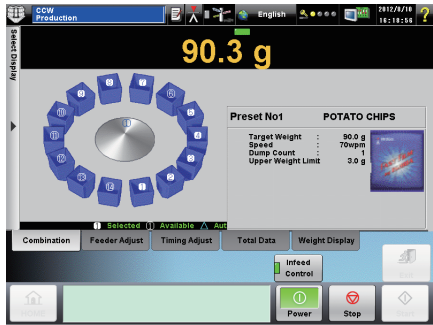


Fig. 4-41

Table 4-2

No.	Name	Function
1	[Combination Display] key	Selects the Combination Display screen.  Fig. 4-42
2	[Expansion Display] key	Selects the Expansion Display screen for the combination weight.  Fig. 4-43

4.6 Temporary Stop and Restart

To stop the production temporarily and then restart it, follow the procedures below.

4.6.1 Temporary Stopping

Stop the production temporarily.

1. Press the [Stop] key.
 - ▶ The [Stop] key is greyed out and the [Start] key appears.
 - ▶ The production is temporarily terminated.

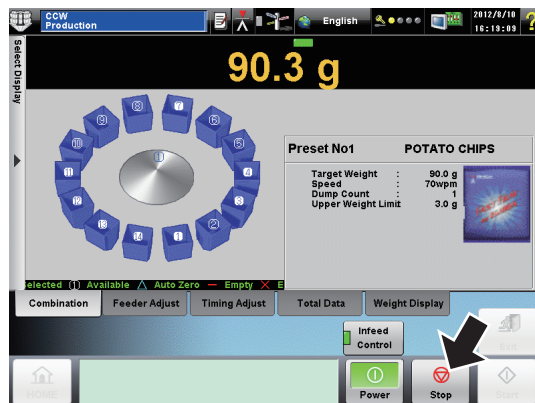


Fig. 4-44

4.6.2 Restarting

Restart the production.

1. Press the [Start] key to restart the production.
 - ▶ The [Start] key is greyed out and the [Stop] key appears.
 - ▶ The production restarts.

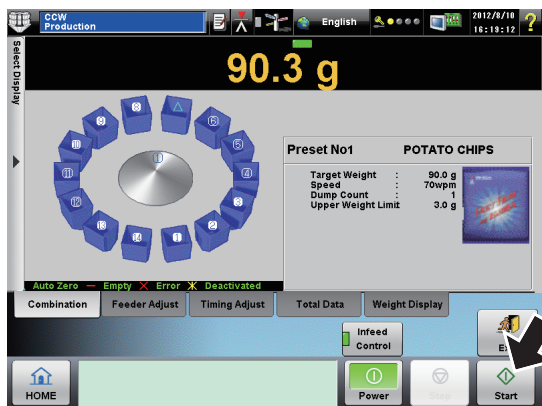


Fig. 4-45

4.7 Outputting the Operational Status

During the operation, every combination result or current total result can be output as necessary.

4.7.1 Every Combination Output

NOTE

- Every combination output should be set by the site engineer or higher level personnel.

This setting outputs the combination weight value during the production.
To perform every combination output, follow the procedures below.

<Operation procedure>

- In the Production screen, press the [Total Data] tab.
► The Total Data screen appears.



Fig. 4-46

- Press the [Select Total] pop-up key.
► The Select Total pop-up menu appears.



Fig. 4-47

3. Press the [Total Setting] key.

► The Total Setting screen appears.

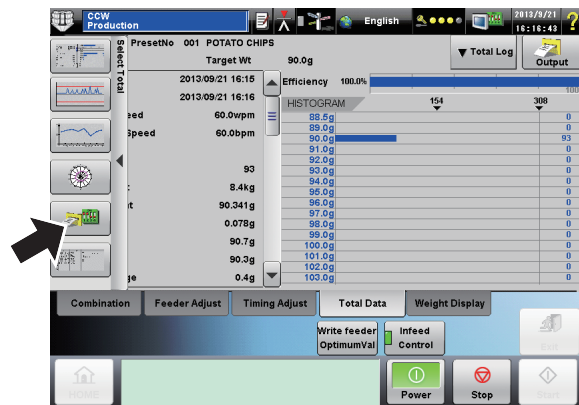


Fig. 4-48

4. Select [On] for the [Every Combination Output] option.

NOTE

- With this setting, the results are output in every two weighings. To cancel this setting, select [Off] for the [Every Combination Output] option.



Fig. 4-49

NOTE

- This option is set to [Off] once the power supply to the device is turned off.

4.7.2 Current Total Output

This setting outputs the total data of the currently selected product to a printer or as a file. To perform the current total output, follow the procedures below.

NOTE

- Printing or file output can be set by the "Installation" or higher level personnel via the Destination ID Tab Screen of the Control Panel screen (Refer to "6.3.5.3 Destination ID Tab Screen").
- The file is output in the text file format and can be opened with the text editor, etc.

<Operation procedure>

1. In the Production screen, press the [Total Data] tab.

► The Total Data screen appears.



Fig. 4-50

2. Press the [Output] key.

► Output starts.



Fig. 4-51

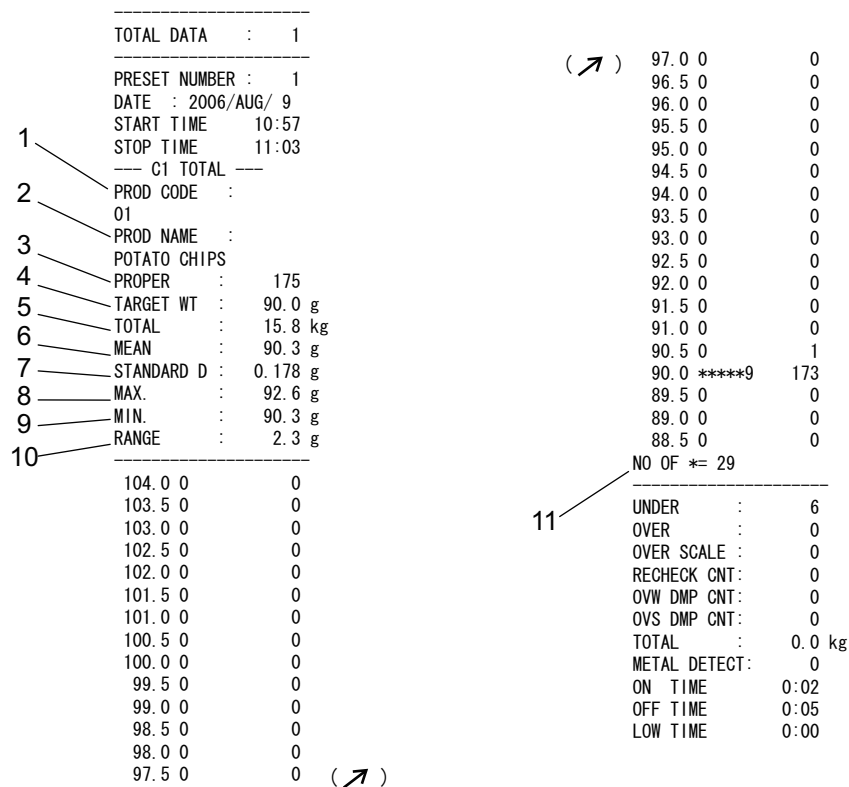


Fig. 4-52

Table 4-3

No.	Name	Remarks
1	Product code	—
2	Product name	—
3	Proper weight	Number of times that the proper discharge is performed.
4	Target weight	—
5	Total value	Total discharged weight.
6	Mean value	—
7	Standard deviation	—
8	Maximum weight	—
9	Minimum weight	—
10	Weight range	(Maximum value)-(Minimum value)
11	Quantity per * mark	Weighing results are displayed in order of weight value, histogram, and number of weighing times from the left. The number displayed in the histogram indicates a percentage of * (e.g. number $\times$ 10%). If a value of * = 18, when the number of weighing times is 45, it will be displayed as "***5" (18 + 18 + 18 $\times$ 50%).

4.8 Loading the Printer Paper

When the message "No Paper" is displayed on the lower part of the screen, follow the procedures below to load the printer paper.

1. Slide the printer cover retainer to open the cover.



Fig. 4-53

2. Pull the printer unit fixed lever to lower the printer unit.

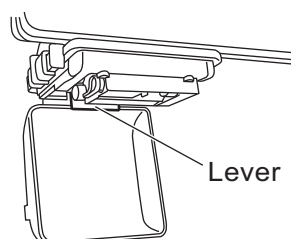


Fig. 4-54

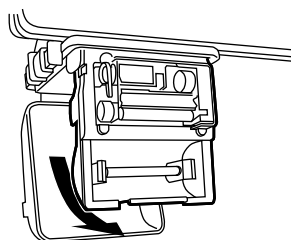


Fig. 4-55

3. Push up the printer head lever.

► The printer head uprears.

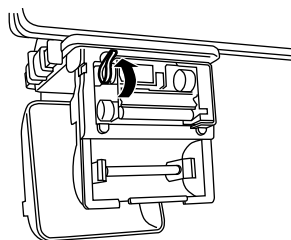
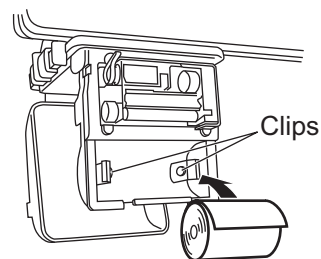
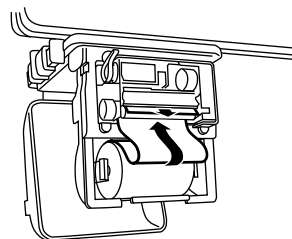


Fig. 4-56

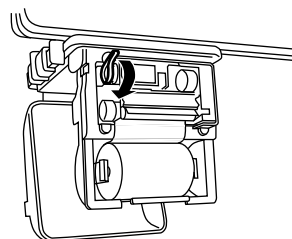
4. Press and open the clips to remove the remaining roll core.
5. Align the roll as shown in the figure and push onto the clips.

**Fig. 4-57**

6. Insert the end of the roll paper into the printer.
7. Advance the paper until the end of the paper comes out about 1cm from the printer head.

**Fig. 4-58**

8. Pull down the printer head lever.
▶The printer head lowers.
9. Push up the printer unit.
10. Close the printer cover.
▶Loading of paper to the printer is completed.

**Fig. 4-59**

4.9 Handling Drive Unit

This section describes how to handle the drive unit.

CAUTION

- **DO NOT, UNDER ANY CIRCUMSTANCES, CLOSE THE HOPPERS WITH HANDS OR TOOLS AS SHOWN IN THE FIGURE. Doing so may damage the drive unit.**

(Left figure shows the weigh hopper and right figure shows the pool hopper.)

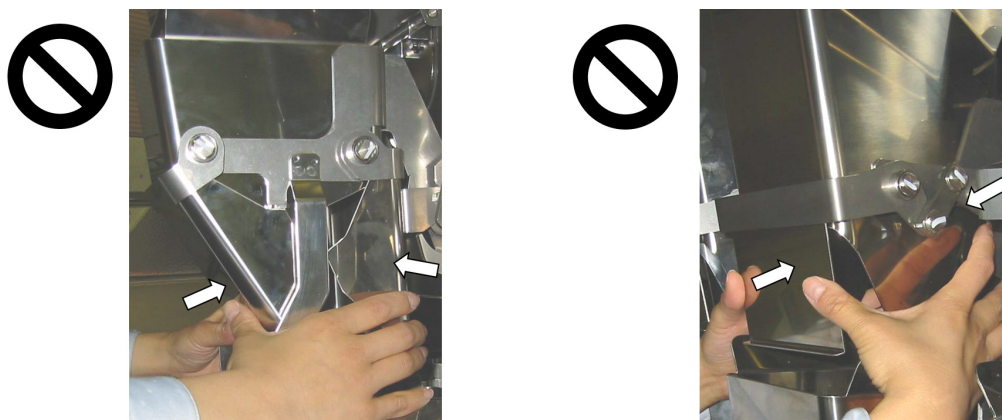


Fig. 4-60

This figure is only for explanation. Your hopper may look different from the above figure.

CAUTION

- **If the main power switch is turned OFF while the hoppers are open, or the hopper remains open due to an electrical power failure, follow the procedures described below to restore.**

1. Turn on the main power switch.
2. Display the Full Open Lock screen (Refer to "6.8 Full Open Lock Screen").
3. Select the relevant weigh hopper and pool hopper, and perform the Full Open Lock operation.
4. Close the open hoppers.
5. Make sure that all of the hoppers are closed.
6. If any of the hoppers remain open, repeat steps 3 to 5.